

Deep Learning Based on Recurrent Networks: Application to Functional Diversity

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SUMMARY

Over the past decade, a wide range of artificial intelligence techniques has emerged and been applied to automatic speech recognition. Numerous deep neural network models, often trained on proprietary datasets, have been developed and published. This master's thesis investigates several recurrent deep learning network models for audio-to-text transcription. The goal of this work is to lay the groundwork for a prototype designed to support individuals with disabilities in academic environments.

Key words: Deep learning – recurrent neural networks – automatic speech recognition – transfer learning – Spanish language datasets.

1 INTRODUCTION

In recent years, research has highlighted how speech recognition systems can serve as highly effective assistive technology tools. While this type of technology can benefit anyone, it offers even greater advantages to individuals with disabilities affecting hearing, writing, attention, and more. Some studies have even explored the use of speech recognition systems as rehabilitation tools for people with language disorders [1].

A study examining the level of inclusion within the Spanish University System, which surveyed 55 universities in Spain, found that approximately 1.7% of students have some form of disability [2]. Back in 2000, St. Mary's University in Canada carried out a three-year initiative known as the Liberated Learning Project, which was evaluated by students with various disabilities. The project involved providing real-time transcriptions of lectures using a specialized version of IBM's ViaVoice that projected live transcripts of the professor's speech [3]. The study concluded that displaying real-time transcriptions could offer benefits to both students with and without disabilities, though it identified transcription accuracy as the main challenge [4].

In recent years, notable advances have significantly improved the accuracy of these systems. With the rise of deep neural networks, research at the intersection of automatic speech recognition and deep learning has flourished. Many of the most influential models have been trained on proprietary datasets, which is why this project contributes to the broader effort of developing open models using open data. The ultimate goal of this project is to build a highly accurate Spanish-language model using publicly available data and deep learning-based approaches.

2 AUTOMATIC SPEECH RECOGNITION

Automatic speech recognition can be described as follows: given an input of audio samples x from a recorded speech signal, apply a function f to map it to a sequence of words P that represents the transcription of what was said.

$$P = f(x) \quad (1)$$

Identifying such a function is a highly challenging task, requiring multiple modeling steps to generate the correct word sequence [5]. The traditional approach has involved extracting features from audio files through a series of preprocessing algorithms. Once audio features such as MFCCs (Mel-frequency cepstral coefficients) are obtained, additional post-processing steps are needed, including acoustic modeling, pronunciation dictionaries, and language models [6]. With the rise of deep learning, end-to-end systems have emerged that eliminate the need for separate acoustic models and pronunciation dictionaries, although they still depend on a language model.

2.1 Error Rates

WER (word error rate) is a metric that measures error based on Levenshtein distance [5]. It can be calculated as follows:

$$WER = \frac{S + D + I}{N} = \frac{S + D + I}{S + D + C} \quad (2)$$

A metric similar to WER is CER, which performs the calculation at the character level instead of the word level. CER uses the same formula as WER but applies it to characters. For both WER and CER, S is the number of substitutions, D the number of deletions, I the number of insertions, C the number of correct words or

characters, and N is the number of reference words or characters [5].

To calculate accuracy, one simply subtracts the WER or CER from 100. WER is more commonly used than CER, so in this work, metrics will be evaluated using WER.

$$\text{Precision} = 100 - \text{WER} \quad (3)$$

2.2 Language Model

A language model plays a crucial role in improving a system’s predictions by enhancing word accuracy and functioning as a form of spell correction during the decoding phase, addressing phonetic errors. For this project, the KenLM toolkit [7] was used to construct the language model, as it allows for the efficient creation of n-gram models from a given dataset. The language model can be derived from the transcripts within the training dataset; however, leveraging a larger external corpus is generally recommended, as it tends to yield superior performance.

3 DATASETS

Most speech recognition models developed by major companies are trained on proprietary datasets. It is estimated that achieving strong performance requires training models on more than 10,000 hours of audio paired with corresponding transcriptions. For instance, in the research paper Deep Speech 2 [8], the authors trained their model on 12,000 hours of audio, emphasizing that increasing the amount of training data leads to a substantial reduction in error rates. The precise volume of training data required can vary depending on the complexity of the target language, with more complex languages generally demanding larger datasets. Fortunately, a variety of Spanish-language datasets are freely available, as summarized in Table 1.

LibriVox is a volunteer-driven project in which participants record public domain audiobooks [9]. Several datasets have been compiled from these recordings; for example, the Linguistic Data Consortium has released a 73-hour collection, while another compilation available on Kaggle offers 120 hours of audio [10]. For this master’s thesis, the 120-hour Kaggle dataset was selected, featuring recordings of 17 different speakers who read a variety of books.

Common Voice, a Mozilla initiative, invites volunteers to record audio from provided text or to validate the recordings of others [11]. The Spanish-language subset of Common Voice includes over 167 validated hours of audio and features contributions from 8,252 unique speakers.

VoxForge provides a dataset comprising 53 hours of transcribed audio. To create the Spanish-language version, a script was adapted to compile and process the relevant data [14].

Tatoeba is a project focused on collecting recordings of common phrases in various languages [12]. For this dataset, audio files and their transcriptions were extracted from the Tatoeba website and subsequently published on Kaggle [13]. The resulting Spanish-language dataset consists of 55 hours of recordings.

Dataset	Hours
Librivox Kaggle	120
Common Voice Spanish	167
VoxForge	53
Tatoeba	55

Table 1. Spanish language datasets

Each dataset has different characteristics. Common Voice features a greater diversity of voices and, therefore, a wider range of accents, microphones, and acoustic environments with more background noise. The audio recordings in LibriVox, VoxForge, and Tatoeba were made more carefully and have less diversity in accents, microphones, and acoustic environments.

3.1 Gathering More Data

While several useful datasets are publicly available, the volume of data they offer is not comparable to the vast amounts used by large companies. In this work, two approaches are proposed to rapidly and cost-effectively expand the dataset. The first approach involves extracting transcripts from YouTube videos, while the second focuses on generating synthetic audio.

3.1.1 Youtube

Platforms like YouTube host a vast number of videos with transcriptions. Many of these videos are published under the Creative Commons license, which permits their use for training models. In this project, a script was developed that, given a YouTube video, can extract both the audio and the transcription [14]. However, this approach was ultimately not pursued, as the transcriptions may be of low quality and would require hours of validation.

3.1.2 Synthetic Data

Synthetic audio data was also generated using public domain texts. These texts were processed through AWS Polly [15], a service that converts text to speech using synthetic voices produced by either concatenative synthesis models or neural models. This process yielded a synthetic dataset of approximately 60 hours. Four female voices and two male voices were employed, representing accents from Mexico, Spain, and the United States [16].

To evaluate the characteristics of the synthetic audio, an analysis was performed using the same transcript to compare synthetic and real audio samples. As illustrated in Figures 1 and 2, although both types of audio may contain identical spoken content, they can exhibit notable differences in their frequency distributions.

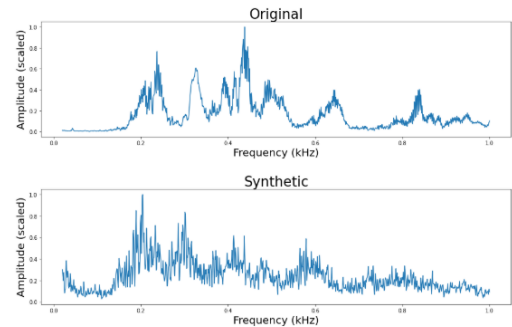


Figure 1. Frequency comparison of an audio file generated synthetically and the original audio both using the same transcription.

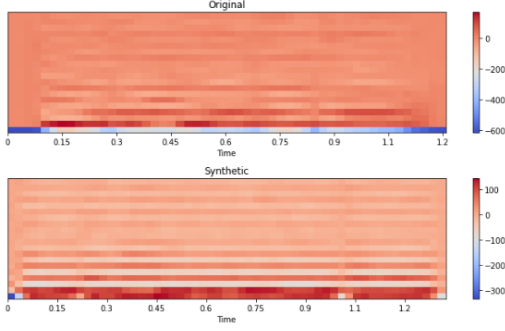


Figure 2. Spectrogram comparison of a synthetically generated audio and the original audio with the same transcription.

3.2 Data Cleaning

For the data cleaning process, an alphabet was defined to exclude any elements containing characters not listed in the provided alphabet text file. In this case, the alphabet encompassed all letters used in the Spanish language. Additionally, a data cleaning tool called Deep Speech Cleaner [17] was employed to further refine the dataset. This tool performs substitutions within the transcriptions and removes silence from the beginnings and ends of the audio files.

4 MODELS

In recent years, numerous models have been introduced that have significantly advanced the field of automatic speech recognition, many of which leverage deep learning techniques. For example, Facebook’s wav2letter employs convolutional neural networks (CNNs), offering considerably shorter training times compared to other architectures. In contrast, models such as Deep Speech and Deep Speech 2 are distinguished by their use of recurrent neural networks (RNNs) within their architectures. Similarly, models like ESPnet also rely on RNNs, though they adopt a different approach from that of Deep Speech. The model developed in this work is based on the Deep Speech 2 architecture.

4.1 Deep Speech

In 2014, Baidu introduced the Deep Speech model, which outlined the implementation of a speech recognition system using recurrent neural networks (RNNs) [18]. At the time, the results achieved were state-of-the-art, and subsequent research further enhanced the model’s efficiency and performance [8]. Today, an active and growing community continues to advance these models, leading to various Deep Speech implementations built on different deep learning frameworks such as TensorFlow, PyTorch, and PaddlePaddle. For this project, the TensorFlow-based implementation maintained by Mozilla was selected [19].

The original Deep Speech architecture comprises three main components: an input layer that processes spectrogram data, a bidirectional long short-term memory (LSTM) layer, and a softmax output layer that produces the final classification predictions [18]. This architecture was later extended in the Deep Speech 2 model,

which features a deeper structure incorporating one to seven recurrent layers along with one to three convolutional layers, as illustrated in Figure 3. Additionally, batch normalization is applied after each layer to improve training stability and convergence [8].

A central element of these architectures is the use of Connectionist Temporal Classification (CTC). CTC is an algorithm designed to address the challenge of unknown alignment between input and output sequences. In the context of automatic speech recognition, CTC enables the mapping of characters from a transcript to their corresponding audio segments by using a beam search decoding algorithm. A language model is then integrated into the system, effectively serving as a spell checker to further enhance inference accuracy [20].

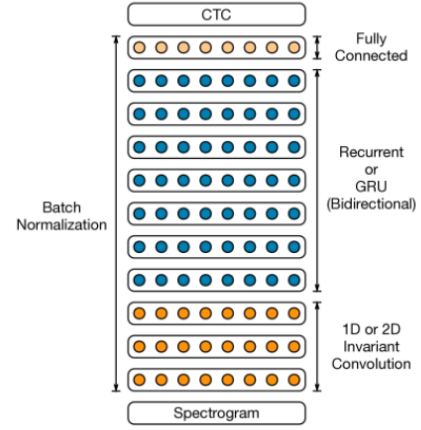


Figure 3. DeepSpeech 2 architecture.

4.2 Language Model Development

To develop the language model, the Spanish Billion Word Corpus was employed. This corpus contains approximately 1.5 billion words drawn from a range of sources, including Wikipedia, United Nations documents, and the OPUS project, among others. In total, the corpus comprises 1,420,665,810 words, 46,925,295 sentences, and 3,817,833 unique tokens [21].

The KenLM toolkit, an open-source program widely used by researchers for building language models, was utilized for this project [7]. To determine the optimal number of n-grams, several models were constructed and evaluated. The model that produced the lowest word error rate (WER) on the Common Voice dataset was selected. As shown in Table 2, the 4-gram model achieved the best performance.

N-grams	WER	CER
(No language model)	1.000	0.625
3-grams	0.429	0.230
4-grams	0.410	0.223

Table 2. Language model comparison

5 TRANSFER LEARNING

Training a speech recognition model from scratch typically demands substantial resources, including time, funding, computational hardware, and large volumes of data. To address these chal-

lenges, this project explored various experiments using the concept of transfer learning. Transfer learning is a machine learning technique designed to improve a model's performance by leveraging knowledge gained from training on data from other domains. A specific form of transfer learning applied here is known as model adaptation, which involves training a model in one language and subsequently retraining portions of that model in another language. In this context, a pre-trained model with access to extensive English-language data can be adapted and retrained for Spanish to achieve improved results [22].

Mozilla's implementation of Deep Speech facilitates model adaptation by enabling the selective removal of certain layers from a pre-trained model, initializing new layers with new data, combining these with the retained pre-trained layers, and updating all layers through the gradient descent algorithm. Crucially, the final layer of the pre-trained model must be removed and reinitialized so that it can operate with the new target alphabet [23].

6 EXPERIMENTS

Two different computing environments were utilized to conduct the experiments. Initially, several experiments were run on the Google Colaboratory Pro platform, which provides access to Nvidia Tesla P100 GPUs. While some experiments completed successfully under this setup, the increasing volume of training data led to extended training times, with some models requiring up to 5–7 days to train. To address this limitation, an alternative environment was established using a machine equipped with an Nvidia Tesla V100 GPU on the Google Cloud Platform. This transition significantly accelerated the experimentation process, enabling more efficient model training and evaluation.

6.1 Hyperparameters

For most experiments, a learning rate of 0.001 was used with a hidden layer size of 2048. Various dropout values were tested, ranging from 0.10 to 0.30. Most of the final models use a dropout rate of 0.22. The number of epochs was set to 30, but due to the use of early stopping, the models typically stopped between 6 and 15 epochs. The beam width parameter was kept at 1024. All other hyperparameters were set to their default values. As an illustration, Figure 4 shows the loss function using the hyperparameters described above for the model trained with the librivox + tatoeba + voxforge dataset.

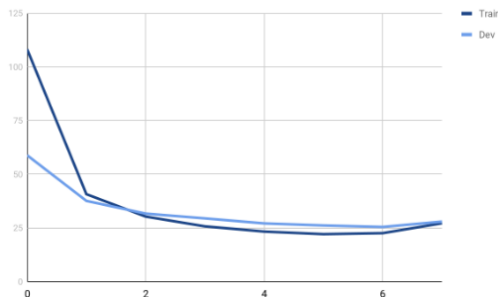


Figure 4. Loss function for the training and test sets using the librivox–tatoeba–voxforge dataset over 7 epochs.

6.2 Training

The first experiment involved training a model using the Common Voice dataset, which consists of over 167 hours of audio. The model initially achieved a WER of 1.0 since no language model was used. After developing and applying the proposed language model, the WER dropped to 0.429 with the 3-gram decoder and to 0.410 with the 4-gram decoder (see Table 2).

Next, models were developed using the Common Voice, LibriVox, VoxForge, and Tatoeba datasets independently, but all were evaluated using the same Common Voice test set. The model trained with LibriVox data achieved a WER of 0.7, while Tatoeba reached 0.85, VoxForge 0.93, and Common Voice 0.410, as shown in Figure 5.

In the second round of experiments, a model was trained using various combinations of training and test datasets. The results are shown in Table 3. First, the model was trained on the LibriVox dataset and evaluated with different test sets from the Common Voice, VoxForge, Tatoeba, and LibriVox datasets. For the Common Voice test set, the model achieved a WER of 0.694; VoxForge yielded a WER of 0.603; Tatoeba, 0.324; and LibriVox, 0.104.

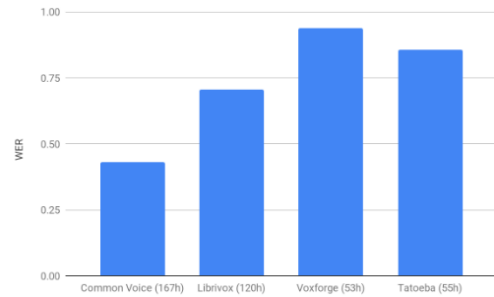


Figure 5. WER using various models trained with different datasets and evaluated with the Common Voice test set.

Subsequently, the LibriVox, Tatoeba, and VoxForge datasets were combined—amounting to approximately 228 hours of audio—for training, and the model was evaluated using test sets from all datasets. The LibriVox test set achieved a WER of 0.138 with the 3-gram language model and 0.132 with the 4-gram model. Tatoeba achieved 0.188, and VoxForge 0.462. Finally, all datasets were combined for training, and the model was evaluated, resulting in a WER of 0.505 for Common Voice, 0.432 for VoxForge, 0.252 for Tatoeba, and 0.151 for LibriVox. When using a mixed test set (LibriVox + VoxForge + Tatoeba + Common Voice), the WER obtained was 0.309

Test dataset	Librivox	Librivox + Tatoeba + VoxForge	Common Voice + Librivox + Tatoeba + VoxForge
Common Voice	0.694	0.662	0.505
VoxForge	0.603	0.462	0.432
Tatoeba	0.324	0.188	0.252
Librivox	0.104	0.132	0.151
Mixed	0.431	0.362	0.309

Table 3. WER obtained using different training and test sets

The next round of experiments involved adding the synthetic

data to the combined training set of Common Voice + LibriVox + Tatoeba + VoxForge. In the initial attempt, the loss function remained constant, indicating that the model was not learning. As a result, the dropout parameter was lowered from 0.22 to 0.12, and the batch size was reduced from 40 to 20. The results of this model were compared to those of the model that did not use synthetic data, as shown in Figure 6. As can be seen, the use of synthetic data only led to an improvement on the Tatoeba test set; for the rest of the test sets, no benefit was observed.

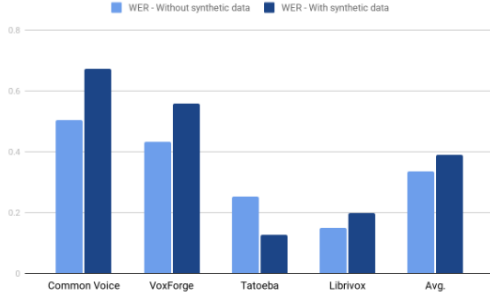


Figure 6. Model comparison using synthetic dataset.

6.3 Transfer Learning

The transfer learning technique was then applied using Mozilla's English-language model trained on 3,260 hours as the base. To do this, the final layer of the English-trained model was removed and retrained using the combined training set of LibriVox + Common Voice + Tatoeba + VoxForge. A learning rate of 0.0001 was used, while the remaining hyperparameters were kept the same.

The language model was also modified, as the version of DeepSpeech that supports transfer learning requires a scorer file to be generated. This scorer file consists of the language model and a trie that contains all the words in the vocabulary.

The results were remarkable across all test sets. The WER on the Common Voice test set dropped from 0.505 to 0.266, on VoxForge from 0.432 to 0.264, on Tatoeba from 0.252 to 0.130, and on LibriVox from 0.151 to 0.075. Using a mixed test set formed from all datasets (LibriVox + VoxForge + Tatoeba + Common Voice), the WER dropped from 0.309 to 0.167.

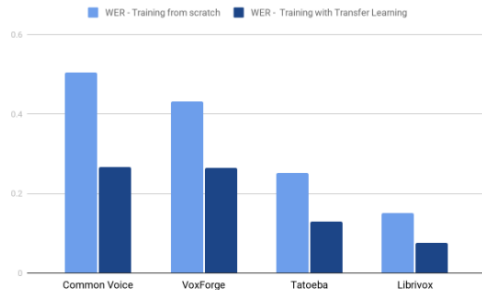


Figure 7. Comparison between model trained from scratch and model using transfer learning.

6.4 Testing using VAD Streaming

For the final prototype test, the final model was tested in real time using both the computer's built-in microphone and a lavalier mi-

crophone. Tests showed that the lavalier microphone may be a better option, as it more effectively blocks background noise and produces clearer audio. To adapt the model, a script developed by Mozilla called VAD Streaming [24] was used. To run the model in real time, it is necessary to import both the model and the scorer file into the script provided by Mozilla. These files can be downloaded from the repository for this master's thesis [14].

6.5 Analysis

Figure 8 shows a summary of the models' performance. The model that achieved the best results was the transfer learning model, as the error rate improved significantly across all test sets. The only test set where the transfer learning model was not superior was Tatoeba, where the model using synthetic data showed a slight improvement compared to the transfer learning model.

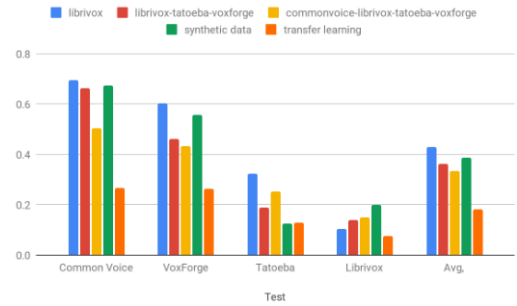


Figure 8. Comparison of all models with different test sets. The best performance was achieved by the model using transfer learning.

The transfer learning model shows highly competitive accuracy on most audio samples. The following table presents some test cases where the accuracy reached 100%.

Original	Result
estas inestables discusiones fronterizas causaron mucha tensión cada uno de ellos es un derecho exclusivo sujeto a ciertas limita- ciones la inhalación del vapor puede originar edema pulmonar	estas inestables discusiones fronterizas causaron mucha tensión cada uno de ellos es un derecho exclusivo sujeto a ciertas limita- ciones la inhalación del vapor puede originar edema pulmonar

Table 4. High-performing results where the transcriptions were 100% accurate.

To better understand the model's errors, the test sets with the worst performance were examined. It was observed that the model struggles to accurately transcribe proper names in other languages as well as words from foreign languages. Table 5 shows some examples where the model performed poorly.

Original	Result
akiko kimura kazuya kawabata eztute oriyec cambiaticu gure jaungoicaren itzá	aquí co que mora es suya que gobat este origen cambia tuco gure angoy caen y

Table 5. The model has problems transcribing names and words in other languages

Table 6 presents a comparison of WER with other models published in the literature. Mozilla’s English-language model achieved a WER of 0.07 on the LibriSpeech test set, which is equivalent to the WER obtained using the LibriVox test set in Spanish. When using a mixed test set composed of all datasets (LibriVox + VoxForge + Tatoeba + Common Voice), the WER of the Spanish model is 0.16—an error rate that is comparable with the results of other models.

Language	Dataset	Hours	Test dataset	WER
English (Baidu)	Switchboard, Fisher, WSJ	7380	Hub5	0.16
English (Mozilla v0.6.1)	Switchboard, Fisher, LibriSpeech	3260	LibriSpeech	0.07
Russian (Mozilla)	Yt-vad-1k, Yt-vad-600	1650	VoxForge	0.18
German (Mozilla v0.5)	Tuda-DE, VoxForge	162	Tuda-DE, VoxForge	0.15
Italian (Mozilla)	VoxForge, CLIPS, Audiobooks, Common Voice	406	Mixed	0.13
Spanish (This work)	Transfer Learning + all datasets	395	Mixed	0.16
Spanish (This work)	Transfer Learning + all datasets	395	Librivox	0.07

Table 6. Comparison of WER across models

7 CONCLUSION

In this work, recurrent deep learning models were used for speech-to-text conversion with the goal of developing a prototype to support functional diversity. First, an analysis of available Spanish-language datasets was carried out, along with an exploratory study of their characteristics. Due to the scarcity of data, various methods for acquiring additional data were explored. Two new datasets were created and published: the first was extracted from the Tatoeba website, and the second was generated using synthetic data. Additionally, a script was developed to extract audio with transcripts from YouTube videos, and another script was modified to create the VoxForge dataset. Using the collected datasets, several models based on the Deep Speech 2 architecture were analyzed, and the use of transfer learning was explored to improve model performance. The best results were obtained using transfer learning with data from Common Voice, Tatoeba, VoxForge, and LibriVox. The outcomes are comparable to those of models published in other languages, and a functional prototype was developed. The model, scripts, and datasets can be downloaded from the repository associated with this project [14].

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REFERENCES

- [1] Alcántud Marín, F., Dolz, I., Gaya, C., & Martín, M., 2006. The voice recognition system as a way of accessing the computer for people with physical disabilities, *Technology and Disability*, **18**(3), 161–168.
- [2] Comité Español de Representantes de Personas con Discapacidad, 2017. Tercer Estudio sobre Universidad y Discapacidad, Fundación Universia.
- [3] Bain, K., Basson, S. H., & Wald, M., 2002. Speech Recognition in University Classrooms: Liberated Learning Project, ACM Conf. Assistive Technologies.
- [4] Helal, A., Mokhtari, M., & Abdulrazak, B., 2009. The Engineering Handbook of Smart Technology for Aging, Disability and Independence, Wiley.
- [5] Kamath, U., Liu, J., & Whitaker, J., 2019. Deep Learning for NLP and Speech Recognition, Springer.
- [6] NithyaKalyani, A., & Jothilakshmi, S., 2019. Speech Summarization for Tamil Language, in *Intelligent Speech Signal Processing*, 113–138.
- [7] Heafield, K., 2020. KenLM Language Model Toolkit, <https://kheafield.com/code/kenlm/>.
- [8] Amodei, D., et al., 2015. Deep Speech 2, Baidu Research, <https://arxiv.org/pdf/1512.02595.pdf>.
- [9] Librivox, 2020. Free Public Domain Audiobooks, <https://librivox.org/>.
- [10] Fonseca, C., 2020. 120 Hours of Spanish Speech, Kaggle, <https://www.kaggle.com/carlfm01/120h-spanish-speech/>.
- [11] Mozilla, 2020. Common Voice, <https://voice.mozilla.org/>.
- [12] Tatoeba, 2020. Collection of Sentences, <http://tatoeba.org>.
- [13] Sánchez, G., 2020. Tatoeba Spanish Speech Dataset, Kaggle, <https://www.kaggle.com/gabrielsanchez/tatoeba-spanish-speech-dataset>.
- [14] Sánchez, G., 2020. Master’s Thesis Repository, <https://github.com/gabrielsanchez/tfm>.
- [15] Amazon, 2020. AWS Polly, <https://aws.amazon.com/polly/>.
- [16] Sánchez, G., 2020. Synthetic Spanish Speech, Kaggle, <https://www.kaggle.com/gabrielsanchez/synthetic-spanish-speech>.
- [17] Silentrus, 2020. DeepSpeech Cleaner, <https://github.com/silentrus/deepspeech-cleaner>.
- [18] Hannun, A., et al., 2014. Deep Speech, Baidu Research, <https://arxiv.org/abs/1412.5567>.
- [19] Mozilla, 2020. Mozilla DeepSpeech, <https://github.com/mozilla/DeepSpeech>.
- [20] Hannun, A., 2017. Sequence Modeling with CTC, Distil, <https://distill.pub/2017/ctc/>.
- [21] Cardellino, C., 2016. Spanish Billion Word Corpus, <https://crscardellino.github.io/SBWCE/>.
- [22] Kunze, J., et al., 2017. Transfer Learning for Speech Recognition, ACL Workshop.
- [23] Mozilla, 2020. Deep Speech v0.7.3 Documentation.
- [24] Mozilla, 2020. Microphone VAD Streaming, https://github.com/mozilla/DeepSpeech-examples/tree/r0.7/mic_vad_streaming.
- [25] Agarwal, A., Zesch, T., 2019. German End-to-End Speech Recognition, Konvens.
- [26] Weibin, C., 2019. Deep Speech ASR for Italian, MSc Thesis, Politecnico di Milano.
- [27] Graves, A., Mohamed, A., Hinton, G., 2013. Speech Recognition with Deep Recurrent Neural Networks, Univ. of Toronto.
- [28] Meyer, J., 2019. Multi-task and Transfer Learning in Low-resource Speech Recognition, Univ. of Arizona.